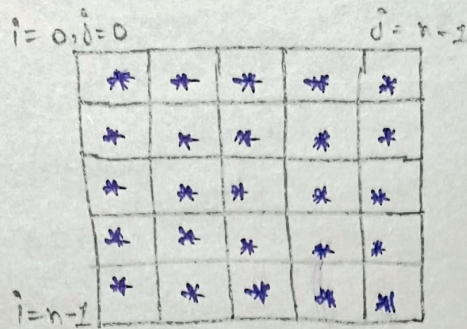


DSA Series :- Module 1

1. Star Patterns -

1. Square :-

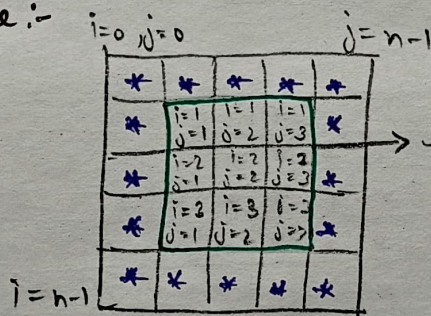
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *



```
⇒ int main() {  
    int n = 5;  
    for (int i = 0; i <= n - 1; i++) {  
        for (int j = 0; j <= n - 1; j++) {  
            cout << "*" << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

2. Hollow Square :-

* * * * *
* *
* *
* *
* * * * *



if ($i=0 || j=0 || i=n-1 || j=n-i$)
 cout << "*" ;
else
 cout << " _ ";

```
⇒ int main() {  
    int n = 5;  
    for (int i = 0; i <= n - 1; i++) {  
        for (int j = 0; j <= n - 1; j++) {  
            if ( $i=0 || i=n-1 || j=0 || j=n-1$ ) {  
                cout << "*" ;  
            }  
            else {  
                cout << " _ ";  
            }  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

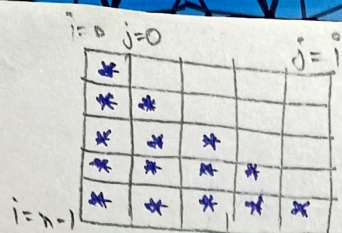

3. Triangle -

```

*
**
***
****
*****

```

⇒



```

int main() {
    int n = 5;
    for (int i = 0; i <= n-1; i++) {
        for (int j = 0; j <= i; j++) {
            cout << " * " << " ";
        }
        cout << endl;
    }
    return 0;
}

```

4. Inverted Triangle - (Reverse approach of upper triangle)

```

*****
****
***
**
*

```

⇒

```

int main() {
    int n = 5;
    for (int i = n-1; i >= 0; i--) {
        for (int j = 0; j <= i; j++) {
            cout << " * " << " ";
        }
        cout << endl;
    }
    return 0;
}

```

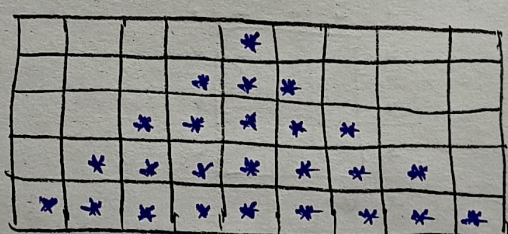
5. Pyramid:-

```

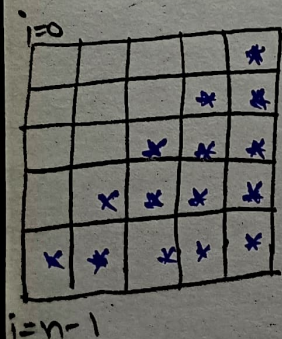
      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *

```

⇒



+ For Left side half of pyramid -



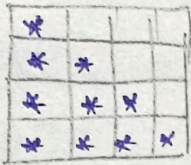
=

```

int main() {
    int n = 5;
    // For Space
    for (int i = 0; i <= n-1; i++) {
        for (int j = 1; j <= n-i-1; j++) {
            cout << " ";
        }
        for (int j = 1; j <= i+1; j++) {
            cout << " * " << " ";
        }
        cout << endl;
    }
}

```


⇒ for Right side of pyramid -



```

for (int i = 0; i <= n-1; i++) {
    for (int j = 1; j <= i; j++) {
        cout << "*" << " ";
    }
    cout << endl;
}

```

⇒ Final :- int main() {
 int n = 5;

// left half pyramid

```

for (int i = 0; i < n; i++) {
    for (int j = 1; j <= n-i-1; j++) {
        cout << " ";
    }
    for (int j = 1; j <= i+1; j++) {
        cout << "* ";
    }
}

```

// right half pyramid

```

for (int j = 1; j <= i; j++) {
    cout << "* ";
}
cout << endl;

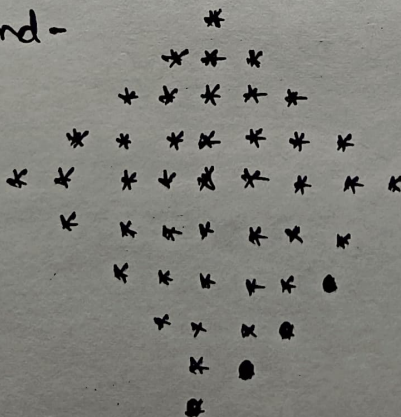
```

```

} return 0;
}

```

6. Diamond -




```
⇒ int main() {
```

```
    int n = 5;
```

```
    // upper half pyramid
```

```
    for(int i=0; i<n; i++){
```

```
        for(int j=1; j<=n-i-1; j++){
```

```
            cout << " ";
```

```
        }
```

```
        for(int j=1; j<=i+1; j++){
```

```
            cout << "* ";
```

```
        }
```

```
        for(int j=1; j<=i; j++){
```

```
            cout << "* ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    // lower half diamond
```

```
    for(int i=n-2; i>=0; i--){
```

```
        for(int j=1; j<=n-i-1; j++){
```

```
            cout << " ";
```

```
        }
```

```
        for(int j=1; j<=i+1; j++){
```

```
            cout << "* ";
```

```
        }
```

```
        for(int j=1; j<=i; j++){
```

```
            cout << "* ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```


7. Half Diamond:-

```
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*
```

⇒

```
int main() {
    int n = 5;

    // upper half diamond
    for (int i = 0; i < n; i++) {
        for (int j = 0; j <= i; j++) {
            cout << " * ";
        }
        cout << endl;
    }

    // lower half diamond
    for (int i = n - 2; i >= 0; i--) {
        for (int j = 0; j <= i; j++) {
            cout << " * ";
        }
        cout << endl;
    }

    return 0;
}
```

8. Half Diamond Inverted -

```
* * * * *
* * * * *
* * * *
* * *
* *
*
```

⇒

```
int main() {
    int n = 5;

    for (int i = n - 1; i >= 0; i--) {
        for (int j = 1; j <= n - i; j++) {
            cout << " ";
        }

        for (int j = 1; j <= i + 1; j++) {
            cout << " * ";
        }

        for (int j = 1; j <= i; j++) {
            cout << " * ";
        }

        cout << endl;
    }

    return 0;
}
```


2. Number Patterns -

1. Square :-

1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1

⇒

```
int main() {  
    int n = 5;  
    for (int i = 0; i < n; i++) {  
        for (int j = 0; j < n; j++) {  
            cout << "1 ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

2. Incrementing :-

1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5

⇒

```
int main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= n; j++) {  
            cout << j << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

3. Right Triangle :-

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

⇒

```
int main() {  
    int n = 5;  
    for (int i = 1; i <= n; i++) {  
        for (int j = 1; j <= i; j++) {  
            cout << j << " ";  
        }  
        cout << endl;  
    }  
    return 0;  
}
```

4. Inverted Diamond -

3 3 3 3 3
4 4 4 4
5 5 5
6 6
7


```

⇒ int int main() {
    int n=5;
    for(int i=3; i<=7; i++){
        for(int j=7; j>=i; j--){
            cout<<i<<" ";
        }
        cout<<endl;
    }
    return 0;
}

```

5. Sandwich Pattern -

```

33333
444
5
444
33333

```

```

⇒ int main() {
    int n=3;
    // upper part
    for(int i=0; i<2; i++){
        for(int j=0; j<i; j++){
            cout<<" ";
        }
        for(int j=0; j<5-2*i; j++){
            cout<<n+i<<" ";
        }
        cout<<endl;
    }
    for(int i=0; i<2; i++){
        cout<<" ";
    }
    cout<<5<<endl;
    // lower part
    for(int i=1; i>=0; i--){
        for(int j=0; j<i; j++){
            cout<<" ";
        }
        for(int j=0; j<5-2*i; j++){
            cout<<n+i<<" ";
        }
        cout<<endl;
    }
    return 0;
}

```


6. Incrementing Diamond Pattern:-

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

⇒

```
int main() {
```

```
    int start = 3;
```

```
    int n = 5;
```

```
    // upper half
```

```
    for (int i = 1; i <= n; i++) {
```

```
        int num = start + i - 1;
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << num << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    // lower half
```

```
    for (int i = n - 1; i >= 1; i--) {
```

```
        int num = start + i - 1;
```

```
        for (int j = 1; j <= i; j++) {
```

```
            cout << num << " ";
```

```
        }
```

```
        cout << endl;
```

```
    }
```

```
    return 0;
```

```
}
```